

everything that is intelligent is smart, but not *vice-versa*. This article covers the use of artificial intelligence (AI) and smart devices in smart homes and smart cities.

Smart home technology is also referred to as home automation that provides convenience, comfort, security and energy-efficiency through the use of smart devices including smart-phones, sensors or other networked devices. The first general-purpose home automation network technology, X10, was developed in 1975. It is a communication protocol for electronic devices that uses electric-powered transmission wiring for signalling and control. A transmitter sends a signal along the house's electric wiring, telling the device to turn on at a specific time.

Household electrical wiring is used to send digital data between X10 devices to control lights and other electrical appliances. Digital data consists of an address and a command sent from a transmitter to a device. This data is encoded onto a 120kHz carrier, which is transmitted as bursts during the relatively-quiet zero crossings of 50Hz or 60Hz AC waveform. One bit is transmitted at each zero crossing. X10 controls in one house do not interfere with X10 controls in a neighbouring house. Inductive filters are used to attenuate X10 signals coming into or going out of the local area.

AI in smart homes

X10 was initially a one-way technology, so while smart devices can take commands, these cannot send data back to the central network. But later, two-way controller devices were introduced that allow for a more robust network. As technology advanced, RF protocols were defined to allow for wireless keypads, remote switches, motion sensors and so on. The wireless protocol operates at a frequency of 310MHz in the US and 433.92MHz in European systems.

While X10 devices are still around, other technologies have emerged to compete for home networking. Instead of going through power lines, many new systems use radio waves to communicate. Now, we are witnessing technologies like the Internet of

Things (IoT), Bluetooth, Wi-Fi and cellphones in home automation.

A wireless network provides more flexibility, convenience and comfort at finger tips. With such emerging technologies as AI, home automation has become smarter and more advanced. Integration of AI has redefined the role of home automation and added new features including home security and energy-efficiency.

How AI-based smart home systems work

All devices in an AI-powered smart home system should be able to communicate with each other and continuously acquire new data for further learning. A truly intelligent smart home device employs the learning system—a multi-layer system capable of making decisions based on historical and real-time data, which requires little to no user control.

For instance, when you leave a room either to go to the bathroom or the balcony, the system should be able to identify your actions and movements, and issue appropriate commands to other devices within the network, either to turn off the lights in that room and turn on the lights in another. The goal of smart home devices is to bring down manual settings to zero. Using AI and machine learning, you can significantly minimise the inconvenience for connected home-owners. Let us take a look at some applications of AI in smart home automation.

Location-based smart devices. Smart home devices can be remotely controlled through smartphone apps and preferences can be auto-managed. For instance, when you are approaching your home and are within a given radius, the system can set up your room with your own preference of lights, curtains and AC temperature. These features add to convenience and comfort.

Voice-enabled devices. Smart devices can be linked with such devices as Google Assistant and Amazon Alexa to provide voice-control options across all home automation systems. Voice technology can be used to control lights, make voice or video calls, read news, place orders, play music and so on. This feature can be used during a medical emergency in notify-



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Model	Range	Test Voltage DC
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MC-903BA	0 - 100 M Ohms	500 V
MC-904BA	0 - 500 M Ohms	500 V
MC-941BA	0 - 1000 M Ohms	500 V
MC-906BA	0 - 200 M Ohms	1000 V
MC-907BA	0 - 500 M Ohms	1000 V
MC-981BA	0 - 2000 M Ohms	1000 V

Digital



DIT 99 Series

Model	Range	Test Voltage DC	Resolution
DIT 99A	0 - 20 M Ohms	100 V	0.01 M Ohms
DIT 99B	0 - 200 M Ohms	250 V	0.1 M Ohms
DIT 99C	0 - 200 M Ohms	500 V	0.1 M Ohms
DIT 99D	0 - 200 M Ohms	1000 V	0.1 M Ohms
DIT 99E	0 - 2000 M Ohms	1000 V	1 M Ohms

5KV



DIT 954

Specification	Test Voltage	Range
Insulation Resistance	1000V / 2500V / 5000V	0.1MΩ to 100GΩ
AC Voltage Measurement	0 - 600VAC (50 - 60Hz)	-
Phase Sequence Test	100V - 450V (Phase - Phase) 40 - 60Hz	-

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